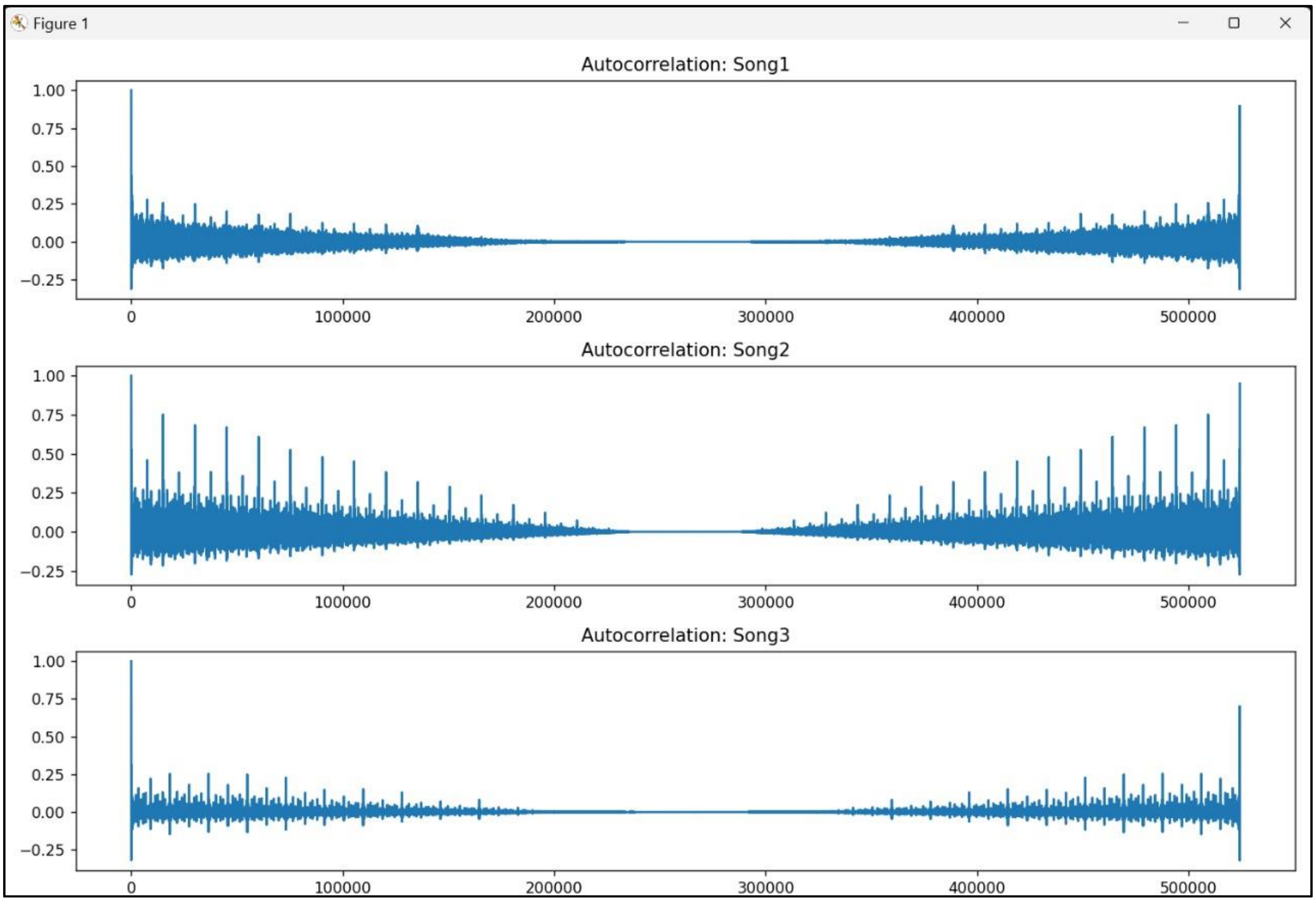
Marwadi University	Marwari University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing(01CT0513)	**Aim(Open Handed)** Simulate Cross Correlation and Auto correlation on Discrete Time Signals.	
Experiment No: 03	**Date: 20-08-2025**	**Enrollment No: 92301733054**

Code:-

```
import matplotlib.pyplot as plt
import numpy as np
import librosa

audio_path1 = "/Users/nikhilbhanderi/Desktop/Experiment 3/Audio/Vande Mataram (Maa Tujhe Salaam)-(Mr-Jat.in).wav"
audio_path2 = "/Users/nikhilbhanderi/Desktop/Experiment 3/Audio/Vande Mataram Karaoke -HQ.wav"

signal1, sr1 = librosa.load(audio_path1, sr=None, duration=60)
signal2, sr2 = librosa.load(audio_path2, sr=None, duration=60)

min_len = min(len(signal1), len(signal2))
signal1 = signal1[:min_len]
signal2 = signal2[:min_len]

def cross_correlation(sig1, sig2):
    return np.correlate(sig1, sig2, mode='full')

def autocorrelation(sig):
    return np.correlate(sig, sig, mode='full')
cross_corr = cross_correlation(signal1, signal2)
auto_corr = autocorrelation(signal1)
lags_cross = np.arange(-len(signal1) + 1, len(signal2))
lags_auto = np.arange(-len(signal1) + 1, len(signal1))
plt.figure(figsize=(12, 6))

plt.subplot(2, 1, 1)
plt.plot(lags_cross, cross_corr, color='b')
plt.title("Cross-correlation (Audio1 vs Audio2)")
plt.xlabel("Lag")
plt.ylabel("Correlation")
plt.grid(True)

plt.subplot(2, 1, 2)
plt.plot(lags_auto, auto_corr, color='g')
plt.title("Autocorrelation (Audio1)")
plt.xlabel("Lag")
plt.ylabel("Correlation")
```

```
plt.grid(True)

plt.tight_layout()
plt.savefig("./Audio_Correlation.png")
print("File Saved at Audio_Correlation.png")
plt.show()
```

Output :-



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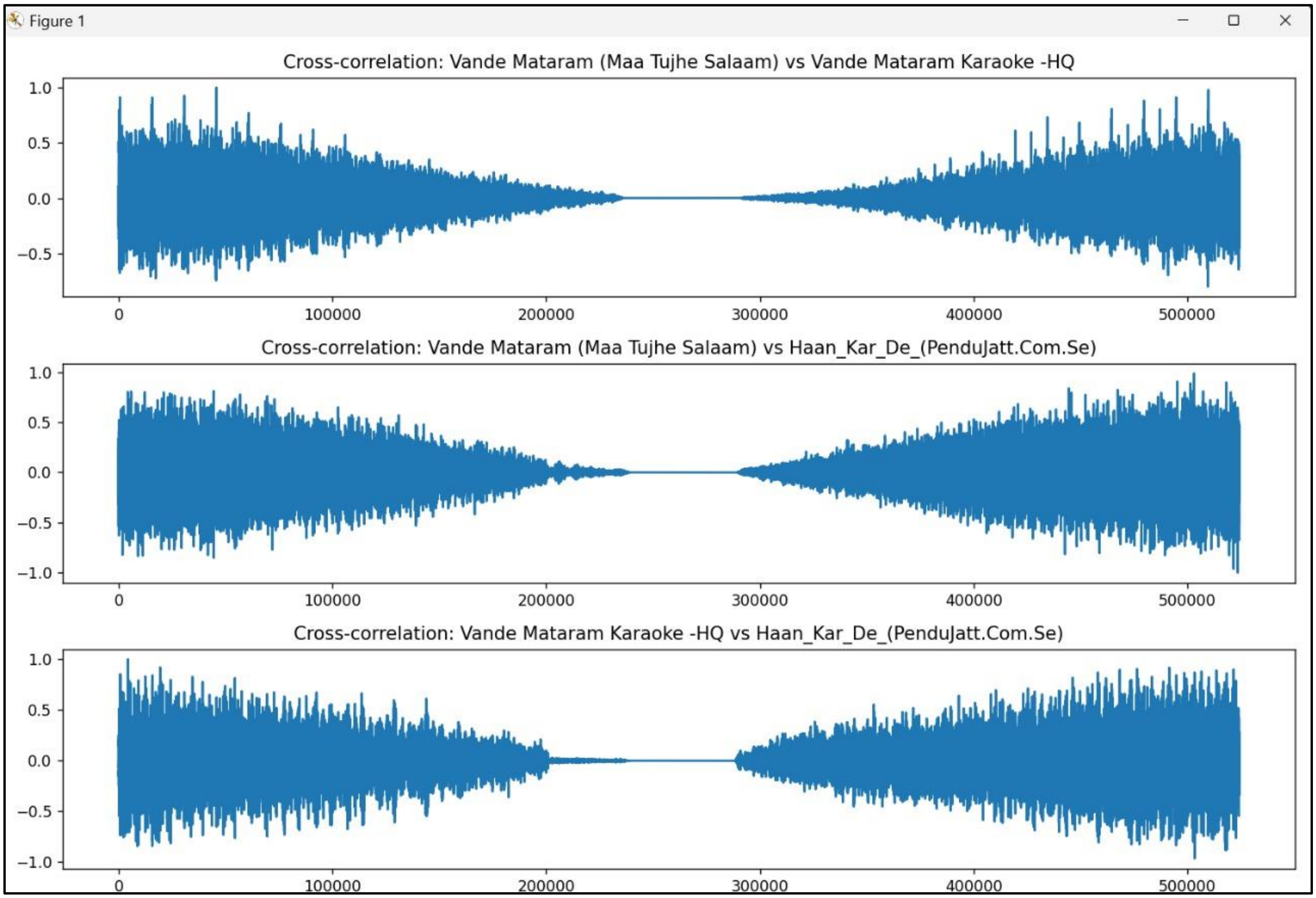

Observation:-

Observations:

1. **Cross-correlation (Audio1 vs Audio2):**
   - The cross-correlation plot shows how similar the original audio and the karaoke version are at different time shifts (lags).
   - Peaks in the cross-correlation indicate points where the two audio signals are most aligned.
   - Since the karaoke track has the same instrumental and timing but no vocals, the cross-correlation is generally high where the instrumental parts match.
   - The maximum peak represents the best alignment or synchronization point between the two tracks.
   - The presence of multiple smaller peaks may indicate sections where rhythms or beats temporarily match.

2. **Autocorrelation (Audio1):**
    - The autocorrelation plot shows the repeating patterns and periodicity within the original audio signal.
    - Peaks at regular intervals correspond to the rhythm or beat of the song.
    - The highest peak occurs at lag 0, which represents perfect self-similarity.
    - Other smaller peaks indicate repeating musical patterns or vocal phrases.
3. **General Insights:**
    - Comparing cross-correlation with autocorrelation helps in understanding both alignment between two tracks and internal repetitive structure of the original track.
    - This analysis can be useful for audio synchronization**,** cover song detection**,** or vocal extraction tasks.

Conclusion:-

We learn this experiment cross-correlation shows similarity between two audio tracks, and autocorrelation reveals repeating patterns within a track.